Data Science

Missing Data



Exploratory Data Analysis

GENERAL ASSEMBLY

The analysis of incomplete data should not be viewed as a technical sideline in applied statistics, but as part of the mainstream. Incompleteness is not a peripheral issue, but one at the core of many statistical analyses.

Donald B. Rubin, Professor of Statistics at Harvard University, and pioneer of multiple imputation methods.

Lesson roadmap

Welcome + Introduction

Defining Missing Data and Its Impact

Types of Missing data

Unit Vs. Item Missingness

Methods for Handling Missing Data

Bring It Home

Learning objectives



1. Summarize the main methods of dealing with missing data and their implications.
2. Explain the difference between unit missingness and item missingness.
3. Understand the main types of missing data.
4. Explain the strategies for imputing missing data and their implications.
5. Handle missing data using **pandas**.

Defining Missing Data and Its Impact

What is 'Missing Data' and why is it important?

Missing data is an ***extraordinarily common*** issue in real-world data sets.

We have missing data whenever, for some variable of interest, no value is stored for it.

Missing data is typically represented as `NaN` (Not a Number), `NULL`, `NA`, or blank cells.

Missing data can **significantly impact analysis results and model performance**!

The proper handling of missing data is something of an art. It is a **critical step in the data preparation process**.

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The Impact of Missing Data

In 1999, **NASA lost $125m** when their climate orbiter rose an error caused by missing unit conversion data.

More locally, **missing patient follow-up data** can drastically undermine the reliability of clinical trials.

Incomplete market data points can dramatically skew investment predictions when financial forecasting, leading to huge losses (and fortunate gains)!



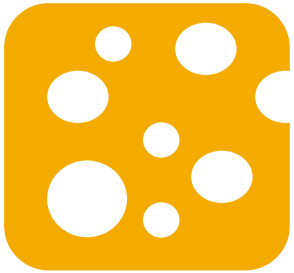




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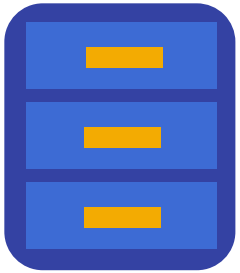
Source 1 Source 2 Source 3

Patterns of Missingness



Swiss Cheese

Random holes perforate the entire data set.



File Cabinet

Entire sections or records missing from the data set.



Diagonal Pattern

Progressive missingness (common in longitudinal data).

Missing Data in Your Work

- [ ] **How has missing data affected *your* work?**

- [ ] Share and compare stories of how missing data has impacted your work.
- [ ] What solutions were offered by yourselves or others, and were they any good?

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Why Data is Missing

Types of Missing Data

1

Missing Completely at Random (***MCAR***)

2

Missing at Random (***MAR***)

3

Missing Not At Random (***MNAR***)

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Missing Completely at Random (MCAR)

MCAR missing data occurs if the **missingness is unrelated to any value in the data set.**

For example, MCAR would occur if lab samples were randomly broken during transport, or if weather sensors malfunctioned occasionally due to random hardware failures.

If missing data is MCAR (and *only* MCAR), *and* is not already biased, then it **does not become biased**. But it **does** lose **statistical power**.

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Missing at Random (MAR)

MAR missing data occurs if the **missingness is related to the observed data, but not to the missing values themselves**.

For example, men may be less likely to report their weight; but this is not related to their actual weight.



Missing at Random (MAR)

To take another example, in financial surveys, higher-income respondents are probably more likely to skip income questions; but we can predict this from their ZIP code, and occupation.

Finally, elderly patients may miss follow-up visits (observable age factor).

MAR missing data may generate bias in your modeling, if not properly handled.



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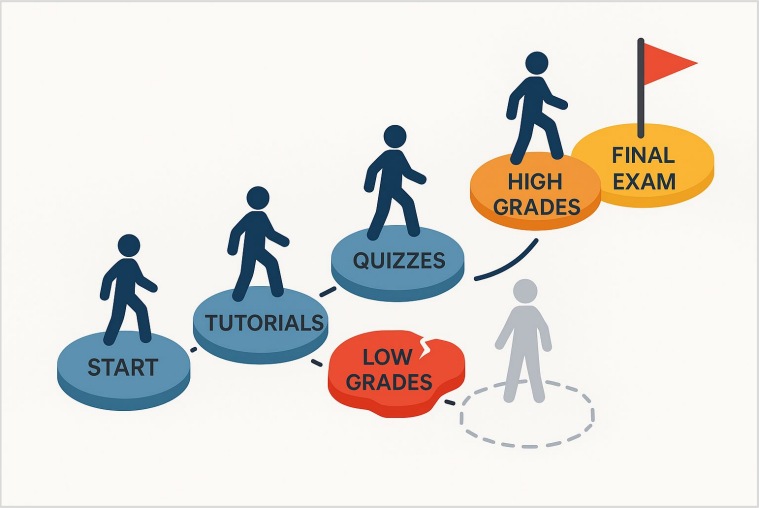
Source: ChatGPT generated image

Missing Not at Random (MNAR)

MNAR missing data occurs if the **missingness is directly related to the missing values.**

For example:

- People with extremely high incomes may be more likely to withhold their income information
- Patients feeling more pain are less likely to complete a pain assessment
- Students performing poorly are less likely to take their final exams



Source: ChatGPT generated image

Missing Not at Random (MNAR)

MNAR missing data can be **extremely problematic** from a modeling perspective.

The missing data mechanism must be carefully modelled (perhaps by using the tools of **causal inference**).



Source: ChatGPT generated image

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Categorising Missing Data

Interestingly, there is **no general procedure** for testing between **MAR** and **MNAR**.

Consequently, to identify the kind of missing data, **subject matter expertise** and **data exploration** are necessary.

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Missing data

Examine the data from the results of a survey on the next slide.

Assess the patterns of missing data and classify each variable's missingness mechanism.

For each variable with missing data, determine whether it's MCAR, MAR, or MNAR and explain your reasoning.

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Full survey results

ID	Age	Income (k)	Education	Job Satisfaction	Health Score	Survey Method
1	25	35	Bachelor	7	85	Online
2	45	85	Master	8	90	Online
3	35	55	Bachelor	5	75	Phone
4	28	42	Bachelor	6	80	Online
5	52	95	PhD	9	95	Phone
6	19	25	High School	3	65	Online
7	31	48	Bachelor	6	78	Online
8	41	78	Master	7	88	Phone
9	29	45	Bachelor	4	70	Online
10	38	62	Master	8	85	Phone
11	24	32	Bachelor	5	72	Online
12	47	88	PhD	9	92	Phone
13	33	51	Bachelor	6	76	Online
14	26	38	Bachelor	4	68	Online
15	44	72	Master	7	86	Phone

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Missing data - what type of missingness is here?

ID	Age	Income (k)	Education	Job Satisfaction	Health Score	Survey Method
1	25	35	Bachelor	7	85	Online
2	45	85	Master	8	NaN	Online
3	35	55	Bachelor	5	75	Phone
4	28	NaN	Bachelor	6	80	Online
5	52	95	PhD	9	95	Phone
6	19	25	High School	NaN	65	Online
7	31	48	Bachelor	6	78	Online
8	41	78	Master	7	NaN	Phone
9	29	NaN	Bachelor	NaN	70	Online
10	38	62	Master	8	85	Phone
11	24	32	Bachelor	5	72	Online
12	47	88	PhD	9	92	Phone
13	33	NaN	Bachelor	6	76	Online
14	26	38	Bachelor	NaN	68	Online
15	44	72	Master	7	86	Phone

- **Income missingness**: is there a pattern between the ages and other characteristics of people with missing income?
- **Job Satisfaction missingness**: what do you notice about the people who didn't report job satisfaction?
- **Health Score missingness**: what survey method was used for the people with missing health scores?

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Missing data - what type of missingness is here?

Income - **MCAR**: The missing income values are all from people with no discernible differences with the other people. The missingness seems random.

Job_Satisfaction - **MNAR**: The missing job satisfaction scores are from people who actually had low satisfaction scores (3, 4, 4). Employees with low job satisfaction may be reluctant to report their true feelings, fearing workplace repercussions or wanting to avoid negative responses. The missingness mechanism depends on the unobserved values.

Health_Score - **MAR**: All 3 missing health scores occur in the "Online" survey group, while all "Phone" respondents provided health scores. This suggests the missingness depends on the survey method (an observed variable), not on the health scores themselves. Online surveys might make it easier to skip sensitive health questions, while phone interviewers ensure completion

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Unit Vs. Item Missingness

Unit Missingness

Unit missingness occurs if an **entire observation (or row) is missing from the data set.**

In this case, the values of all variables for that particular observation are unavailable.



Source

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Unit Missingness: Example

For example, unit missingness occurs if:

- A participant drops out of a longitudinal study
- A survey respondent never returns their questionnaire
- A patient misses a follow-up appointment, or
- A household never responds to a census.

Unit missingness may generate systematic selection bias, provided that certain types of subject are more likely to be missing than others.

Unit missingness is particularly problematic if the mechanism is not MCAR.

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Item Missingness

Item missingness occurs if **specific variables (columns) are missing for observations that otherwise have data.**

Only particular data points are missing, not the entire case.



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Item Missingness: Example

For example:

- A survey participant skips sensitive questions (say about income) but answers other questions, or
- A blood test result is missing for a patient who was otherwise fully assessed.

Item missingness affects specific variables, rather than entire observations.

It can create patterns across variables; creating a ‘swiss-cheese’ pattern in the data set.

Item missingness creates challenges for analyses requiring complete cases.

Comparison of Unit and Item Missingness

Aspect	Unit Missingness	Item Missingness
Scope	Entire observation missing	Specific variables missing
Visualisation	Complete rows missing	Scattered cells missing
Statistical impact	Reduces sample size	Creates imbalance variable coverage
Common causes	Dropout, non-response, attrition	Question-skipping, data collection errors
Typical handling	Weighting, modeling, dropout process	Imputation methods

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The Challenge of Combined Unit and Item Missingness

In many data sets, **both** unit and item missingness occur at once, creating complex patterns:

a. Survey with both non-respondents (unit) and partially completed surveys (item).
b. Clinical trials with patient dropouts (unit) and missed measurements (item).
c. IoT sensors with both complete device failures (unit) and partial sensor failures (item).

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Unit Missingness Vs. Item Missingness

With **unit missingness**, the worry is **representativeness**.

With **item missingness**, the worry is **complication for multivariate analysis**.

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Identifying Missing Data

Let's see it for ourselves!

- [ ] Open **02-missing-data.ipynb, sections 1 and 2**



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Methods for Handling Missing Data

Ideally, find the missing data

If the work is important enough, and the variables are discoverable - for example the revenues of a company, or a ZIP code - then finding the data is the best thing to do.

But what if that's not possible?

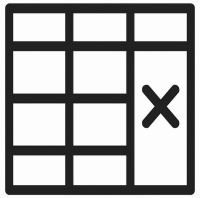


Source

Methods come in three families:

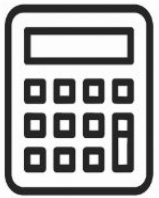
1

Deletion Methods



2

Simple Imputation



3

Advanced Imputation



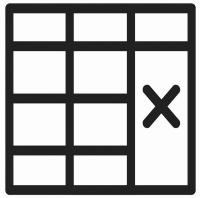
These involve machine learning models and are beyond the scope of this lesson

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Deletion Methods

1

Deletion Methods



Deletion methods involve deleting information within the data set. Let’s look at three:

- Listwise deletion (complete-case analysis)
- Pairwise deletion
- Column deletion

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Listwise Deletion (Complete-Case Analysis)

Method: remove entire observations that have **any** missing values.

Example:

- Customer satisfaction survey analysis may exclude all partially completed responses.
- In healthcare, a clinical trial may include only patients with complete test results.
- In climate analysis, meteorologists may only use weather stations with complete records.

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Listwise Deletion (Complete-Case Analysis)

ADVANTAGES	DISADVANTAGES
Simplicity. Listwise deletion can reduce the scale of a data set in an analytical project.	There is a loss of statistical power. Invisible demographic groups may be systematically removed unbeknownst to the analyst. Statistical bias may be generated, unless the data is MCAR.

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Listwise Deletion (Complete-Case Analysis): Python

```
# Remove all rows with any missing values

df_complete = titanic_df.dropna()
```

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Pairwise Deletion

Method: Use all available data for each calculation separately.

Instead of removing entire cases that have any missing data (listwise deletion), pairwise deletion only excludes cases when they're missing data for the variables currently being used in a specific analysis or calculation.

Example:

- Psychology research may calculate correlations between different personality traits with varying response rates.
- Or in marketing, we may compare product features when customers rated different subsets of features.

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An example of Pairwise Deletion

	Age	Income	Job Satisfaction	Health Score
Mary	25	35,000	7	85
Sayed	45	85,000	8	NaN
Bill	NaN	55,000	5	75
Astrid	28	NaN	6	80
Zara	52	95,000	9	95

If we want to calculate the correlation between 2 features:

- **Age vs Income:** include Mary, Sayed and Zara
- **Age vs Job Satisfaction:** include Mary, Sayed, Astrid and Zara
- **Income vs Job Satisfaction:** include Mary, Sayed, Bill and Zara

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Pairwise Deletion

ADVANTAGES	DISADVANTAGES
Uses more data than listwise deletion.	Can create inconsistent sample sizes across different analyses. Problematic for advanced statistical methods requiring complete correlation/covariance matrices. Most worryingly, it can even produce mathematically impossible results in multivariate analysis.

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Column Deletion

Method: Remove variables (features) with excessive missing values.

Example:

- We drop the optional ‘How did you hear about us?’ field, or ‘years at previous employer’, with 75% missingness.

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Column Deletion

ADVANTAGES	DISADVANTAGES
May delete fewer data points than either previous method.	May remove key explanatory variables. Can significantly affect later research conclusions. Hard threshold decisions have to be made too (is 20% missing excessive? 30%?).

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Column Deletion: Python

```
# Create a copy of the original data set

df_column_deletion = titanic_df.copy()


# Remove columns with more than 30% missing values

threshold = 0.3

df_column_deletion = df_column_deletion.loc[:,
df_column_deletion.isnull().mean() < threshold]
```

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Deletion Methods

Let's see it for ourselves!

- [ ] Open **02-missing-data.ipynb, section 3**



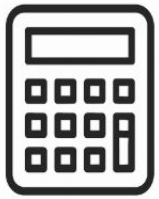
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Simple Imputation Methods

Simple Imputation Methods

2

Simple Imputation



Since each deletion method has its drawbacks, you may resort to simple imputation methods instead: straightforward approaches involving the approximation of the missing values.

Let's look at three:

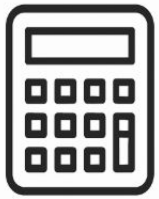
- Mean/median/mode imputation
- Constant value imputation
- Last observation carried forward (LOCF)

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Simple Imputation Methods

2

Simple Imputation



It’s important to note that **it is extremely rare for any of these methods to be adequate for the task.**

It’s important to study them, though, as you will see them frequently in the wild…

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Median/Median/Mode Imputation

Method: Replace missing values with the central tendency measure.

Example:

- In manufacturing, you may find the use of median measurements for missing sensor readings, or
- In healthcare, you might find the use of modal blood type when missing preliminary screening.

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Median/Median/Mode Imputation

ADVANTAGES	DISADVANTAGES
A relatively simple method.	Artificially reduces variance in potentially catastrophic ways. Creates unnatural spikes in distributions. Disrupts natural relationships between variables. Can drastically underestimate standard errors.

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Median/Median/Mode Imputation: Python

```
# Create a copy of the original data set
df_mean_imputed = titanic_df.copy()


# Impute missing values in the 'Age' column with the mean
mean_age = df_mean_imputed['Age'].mean()
df_mean_imputed['Age'] = df_mean_imputed['Age'].fillna(mean_age)


# Check that there are no missing values in the 'Age' column
print(f"Missing values in Age column: {df_mean_imputed['Age'].isnull().sum()}")
```

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Constant Value Imputation

Method: Replace missing values with a predetermined constant.

Example:

- Setting missing customer spending to $0 in retention analysis, or
- Census marking unidentified housing status as ‘unhoused’ for resource allocation.

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Constant Value Imputation

ADVANTAGES	DISADVANTAGES
May be appropriate when missingness has significance.	Introduces artificial data structure, and creates distinctive ‘clusters’ in visualisations and analyses.

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Constant Value Imputation: Python

```
# Create a copy of the original data set

df_constant = titanic_df.copy()


# Impute missing 'Cabin' values with 'Unknown'

df_constant['Cabin'] = df_constant['Cabin'].fillna('Unknown')


# Check the result

df_constant['Cabin'].value_counts()
```

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Last Observation Carried Forward (LOCF)

Method: Replace missing values with the most recent available value.

Example:

- Continuing prior blood pressure reading when a patient misses their latest appointment, or
- Extending previous pollution readings during present sensor outages.

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Last Observation Carried Forward (LOCF)

ADVANTAGES	DISADVANTAGES
Simple to understand and use, particularly in longitudinal studies.	Assumes stability over time (which is often unrealistic). Can dramatically underestimate variability in time series. May hide important trends or events.

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Simple Imputation

Let's see it for ourselves!

- ☐ Open **02-missing-data.ipynb, section 4 onwards**



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DISCUSSIONS: The Methods So Far

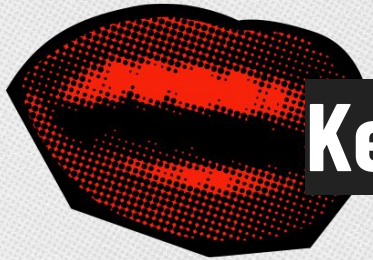
The Methods We've Seen

- **How good have the solutions been that we have seen?**

- Which method is your favourite so far, for handling missing data?
- Can you think of a context in which it struggles to do a good job?
- Can you think of another method that might do better?

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Bring It Home



Key takeaways

1

No perfect solution exists: all methods have trade-offs.

Simple imputation techniques are relatively explicable, but suffer from severe methodological worries.

Advanced ones are harder to explicate, but are more robust.

2

Understanding *why* data is missing is critical.

Whether data are MCAR, MAR or MNAR needs to be ascertained first.

3

There are many ways to handle missing data.

The ideal is to find the data. Otherwise impute. Only delete as a last resort.

4

Missingness can radically affect your inferences.

Both the missingness itself, and your method of handling it, drastically affect the inferences you can validly draw from the data.

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